

MATTHEW SHUNSHI ZHANG

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POSITIONS

C.L.E. Moore Instructor

August 2026 -

MIT, Department of Mathematics, mentored by Prof. Philippe Rigollet.

EDUCATION

University of Toronto

January 2022 - June 2026

PhD, Computer Science, supervised by Prof. Murat Erdogdu. Affiliated with the Vector Institute for Artificial Intelligence. GPA: 4.00/4.00

University of Toronto

September 2020 - January 2022

MS, Computer Science, supervised by Profs. Murat Erdogdu and Animesh Garg. Affiliated with the Vector Institute for Artificial Intelligence. GPA: 4.00/4.00

University of Toronto

September 2016 - May 2020

BASc, Engineering Science.

High Honours, GPA: 3.94/4.00

VISITING POSITIONS

Yale University

January 2025 - April 2026

Hosted by Prof. Sinho Chewi

Georgia Institute of Technology

October 2023 - December 2023

Hosted by Prof. Santosh Vempala

ENS Paris-Saclay

May 2022 - July 2022

Hosted by Prof. Alain Durmus

JOURNAL PUBLICATIONS

[1] Stability of the Kim–Milman flow map

Sinho Chewi, Aram-Alexandre Pooladian, **Matthew S. Zhang**

ECP, 2026

[2] Uniform-in- N log-Sobolev inequality for the mean-field Langevin dynamics with convex energy

Sinho Chewi, Atsushi Nitanda, **Matthew S. Zhang**

SIMA, 2026

[3] Analysis of Langevin Monte Carlo from Poincaré to Log-Sobolev

Sinho Chewi, Murat A. Erdogdu, Mufan (Bill) Li, Ruoqi Shen, **Matthew S. Zhang**

FoCM, 2024

CONFERENCE PUBLICATIONS

[4] Covariance estimation using Markov chain Monte Carlo

Yunbum Kook, **Matthew S. Zhang**

ICML, 2026

[5] Shifted Composition IV: Toward Ballistic Acceleration for Log-Concave Sampling

Jason M. Altschuler, Sinho Chewi, **Matthew S. Zhang**

STOC, 2026

[6] Rényi-infinity constrained sampling with d^3 membership queries

Yunbum Kook, **Matthew S. Zhang**

SODA, 2025

[7] In-and-Out: Algorithmic diffusions for sampling convex bodies

Yunbum Kook, Santosh Vempala, **Matthew S. Zhang**

NeurIPS, 2024 (spotlight)

[8] Sampling from the mean-field stationary distribution

Yunbum Kook, **Matthew S. Zhang**, Sinho Chewi, Murat A. Erdogdu, Mufan Li

COLT, 2024

- [9] **Improved discretization analysis for the underdamped Langevin Monte Carlo**
Matthew S. Zhang, Sinho Chewi, Mufan Li, Krishnakumar Balasubramanian, Murat A. Erdogdu
COLT, 2023
- [10] **Tight regret and complexity bounds for Thompson Sampling via Langevin Monte Carlo**
Tom Huix, **Matthew S. Zhang**, Alain Durmus
AISTATS, 2023
- [11] **Towards a Theory of Non-Log-Concave Sampling: First-Order Stationarity Guarantees for Langevin Monte Carlo**
Krishnakumar Balasubramanian, Sinho Chewi, Murat A. Erdogdu, Mufan Li, Adil Salim, **Matthew S. Zhang**
COLT, 2022
- [12] **Convergence and Optimality of Policy Gradient Methods in Weakly Smooth Settings**
Matthew S. Zhang, Murat A. Erdogdu, Animesh Garg
AAAI, 2022
- [13] **Convergence of Langevin Monte Carlo in Chi-Squared and Rényi Divergence**
Murat A. Erdogdu, Rasa Hosseinzadeh, **Matthew S. Zhang**
AISTATS, 2022
- [14] **One-Shot Pruning of Recurrent Neural Networks by Jacobian Spectrum Evaluation**
Matthew S. Zhang, Bradly Stadie
ICLR, 2020

PREPRINTS

- [15] **Exact simulation of diffusions and improved algorithms for log-concave sampling**
Fan Chen, Sinho Chewi, Alexander Rakhlin, **Matthew S. Zhang**
Preprint, 2026
- [16] **Algorithmic warm-starts for Hamiltonian Monte Carlo**
Matthew S. Zhang, Jason Altschuler, Sinho Chewi
Preprint, 2026
- [17] **Sharp propagation of chaos in Rényi divergences**
Matthew S. Zhang
Preprint, 2026
- [18] **Sublinear iterations can suffice even for DDPMs**
Matthew S. Zhang, Stephen Huan, Jerry Huang, Nicholas Matthew Boffi, Sitan Chen, Sinho Chewi
Preprint, 2025
- [19] **Perspectives on Stochastic Localization**
Bobby Shi, Kevin Tian, **Matthew S. Zhang**
Preprint, 2025
- [20] **Analysis of Langevin midpoint methods using an anticipative Girsanov theorem**
Matthew S. Zhang
Preprint, 2025
- [21] **Benchmarking Model-Based Reinforcement Learning**
Tingwu Wang, Xuchan Bao, Ignasi Clavera, Jerrick Hoang, Yeming Wen, Eric Langlois, **Matthew S. Zhang**, Guodong Zhang, Pieter Abbeel, Jimmy Ba
Preprint, 2019

INVITED TALKS AND PRESENTATIONS

- Midpoint methods for SDEs: Speeding up sampling and generative modelling**
Math for AI (MFAI) workshop, UNAM, Mexico City
April 2026
- Sublinear iterations can suffice even for DDPMs**
Seminar, NUS Computer Science
November 2025
- Perspectives on stochastic localization**
Seminar, University of Pennsylvania
November 2025
- Analysis of Langevin midpoint methods using an anticipative Girsanov theorem**
IMS Young Mathematical Scientists Forum – Statistics and Data Science, NUS
November 2025

Fast and Curious 2: MCMC in action, University of Toronto	September 2025
Wasserstein Gradient Flows in Math and Machine Learning, BIRS	July 2025
Toward ballistic acceleration for log-concave sampling	
Seminar, NUS Mathematics	November 2025
Two Faces of Optimization, ETH	July 2025
Uniform-in-N log-Sobolev inequality for finite-particle systems	
INFORMS 2025	October 2025
Seminar, University of Tokyo	November 2024
Sampling and isoperimetry for finite particle approximations	
SIAM Conference on the Mathematics of Data Science	October 2024
Sampling in the mean-field regime	
Probability Summer School, Saint Flour (Student talk)	July 2024
Seminar, Yale University	March 2024
Isoperimetry and the convergence of LMC	
Machine Learning Summer School, ÉMINES	July 2022
Convergence of LMC in Rényi Divergence	
Applied Mathematics Seminar, CERMICS	June 2022
Analysis of LMC from Poincaré to log-Sobolev	
Complexity of Sampling Working Group, Simons Institute	November 2021

AWARDS

Canada Graduate Scholarship (Doctoral)	2023
University of Toronto Fellowships	2021
Daisy Intelligence Scholarship for Engineering Science	2019
Faculty of Applied Science and Engineering Award	2018
Engineering Society Awards	2018
Jane Elizabeth Ham Scholarship	2017
Canadian Freshman Debating Champion	2017

SERVICE

Conference Reviewer	AISTATS (2022, 2023, 2024, 2025, 2026), NeurIPS (2022, 2023, 2024, 2025, 2026) ICLR (2023, 2025, 2026), ICML (2023, 2024, 2025, 2026), COLT (2025, 2026), ALT (2024, 2025), AAAI (2025, 2026), SODA (2026)
Journal Reviewer	SPA, JAA, FoCM, JMLR, TMLR, Statistica Sinica, IMAJNA, JDS, SINUM
Teaching assistant	CSC412 (Winter 2024), CSC343 (Winter 2023), CSC2532 (Winter 2022), CSC498 (Winter 2021), CSC343 (Fall 2020), ESC180 (Fall 2020)
Organized a reading group on sampling algorithms and stochastic localization at the Georgia Institute of Technology, 2023-2024.	

REFERENCES

Murat Erdogdu, Associate Professor	erdogdu@cs.toronto.edu
Sinho Chewi, Assistant Professor	sinho.chewi@yale.edu
Daniel Lacker, Associate Professor	daniel.lacker@columbia.edu